

Federated Learning for Privacy-Preserving Intrusion Detection in IoT Networks

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Abstract—The expansion of IoT networks, ensuring robust intrusion detection while safeguarding user privacy is a significant challenge. Previous techniques have struggled to reconcile accuracy and privacy. In this research, we offer a unique approach employing federated learning for intrusion detection in IoT contexts. Our technique leverages the Federated Averaging (FedAvg) algorithm. by allowing IoT devices to cooperatively train a global model. To evaluate our technique, we employ the CICIDS 2017 dataset. It simulates realistic network traffic scenarios. Our data demonstrate a remarkable accuracy of 95.2%. It demonstrates the efficiency of federated learning in increasing intrusion detection capabilities. Our investigation reveals encouraging results. Future work intends to solve restrictions, such as broadening the scope to accommodate varied assault situations. Our research illustrates the transformational potential of federated learning in increasing cybersecurity in IoT networks while maintaining user privacy.

Keywords— *IoT networks, leverages the Federated Averaging (FedAvg), privacy Intrusion Detection*

I. INTRODUCTION

The widespread adoption of Internet of Things (IoT) networks has fundamentally transformed multiple facets of contemporary existence. The Internet of Things (IoT) allows for a wide range of applications in several fields [1]. The emergence of IoT networks has raised substantial concerns around security and privacy [3]. The interconnectivity of IoT devices exposes them to a range of security risks, such as unauthorized access and malicious activity [5].

An important obstacle in IoT networks is intrusion detection, which entails the identification and mitigation of illegal access [8]. Conventional intrusion detection systems (IDS) encounter certain constraints when applied to the Internet of Things (IoT) environment. These challenges encompass scalability difficulties and the inability to adjust to dynamic network conditions [9].

This research seeks to investigate the possibilities of federated learning for maintaining privacy while detecting intrusions in IoT networks. The fundamental purpose of this project is to build an effective and privacy-preserving solution for intrusion detection in IoT networks.

The primary aim of this research is to overcome the constraints of conventional intrusion detection techniques in IoT networks. with using federated learning approaches. our objectives are:

- To investigate the feasibility of using federated learning for privacy-preserving intrusion detection in IoT networks.

- To develop a methodology for incorporating federated learning into intrusion detection systems for IoT networks.
- To evaluate the performance of the proposed federated learning-based intrusion detection.

We propose a unique approach combining federated learning with IoT network intrusion detection. This involves data collection and preprocessing. Integrating model training via federated learning, model aggregation, and evaluation. Conduct privacy preservation procedures like differential privacy and homomorphic encryption to protect data. We assess our approach using real-world IoT datasets. contrasting it with established ways to assess effectiveness and efficiency. We strive to enhance the security and privacy of IoT networks while retaining high detection accuracy. Our study contributes to the evolution of intrusion detection techniques in IoT contexts. It is opening the way for enhanced and privacy-preserving safety measures.

II. BACKGROUND

The study on IoT intrusion detection approaches with federated learning provides useful insights into addressing security concerns in IoT contexts. Wheelus and Zhu [14] provide a data-driven protection strategy for IoT network security. It emphasizes the necessity of understanding the dangers and risks. Hassan [15] conducts a comprehensive overview of existing studies on IoT security. That can help highlight major difficulties and study directions. Hamza et al. [16] discuss IoT network security requirements. Threats and responses underline the necessity for effective security mechanisms. Gurunath et al. [17] present an outline of security challenges in IoT networks. The author is highlighting the significance of safeguarding communication connections. Table I briefly summarizes several recent intrusion detection approaches.

TABLE I. SUMMARY OF INTRUSION DETECTION METHODS.

Author(s)	Method	Finding
Dong, B., & Wang, X. [22]	Comparison of deep learning methods	Deep learning methods show promising results compared to traditional methods in network intrusion detection.
Resende, P. A. A., & Drummond, A. C. [26]	Survey of random forest-based methods	Random forest-based methods demonstrate effectiveness in intrusion detection

		systems, offering robust performance.
Govindarajan, M., & Chandrasekaran, R. M. [27]	Neural-based hybrid classification methods	Neural-based hybrid classification methods exhibit potential for accurate intrusion detection in network environments.
Guan, Y., et al. [29]	Y-means clustering method	Y-means clustering method presents a viable approach for detecting anomalies in network traffic.
Kim, G., et al. [30]	Hybrid intrusion detection method	The hybrid method integrating anomaly and misuse detection enhances the accuracy and reliability of intrusion detection.
Oh, S. H., & Lee, W. S. [33]	Anomaly detection method by clustering	Clustering normal user behavior aids in anomaly detection, providing a robust approach for intrusion detection.
Gamage, S., & Samarabandu, J. [25]	Deep learning methods in network intrusion detection	Deep learning methods offer an objective comparison for selecting the most suitable approach in intrusion detection.
Zhu, D., Premkumar, G., Zhang, X., & Chu, C. H. [31]	Data mining for network intrusion detection	Data mining techniques provide valuable insights into network intrusion detection, aiding in effective threat detection.
Ji, S. Y., Jeong, B. K., Choi, S., & Jeong, D. H. [32]	Multi-level intrusion detection method	Multi-level intrusion detection methods enhance the detection capability by identifying abnormal network behaviors.
Tavallae, M., Stakhanova, N., & Ghorbani, A. A. [19]	Evaluation of anomaly-based intrusion detection methods	Anomaly-based intrusion detection methods are essential for identifying novel threats and enhancing network security.

There are traditional intrusion detection methods in IoT networks that can be identified; however, their efficacy contains some limits. [18] Collected by Liu and Lang are works that discuss applications of machine learning and deep learning for intrusion detection systems, which introduce a new technology and could be advantageous in getting more accurate detection. According to Tavallae et al. [19], credibility assessment is a prerequisite to anomaly-based intrusion detection, with well-defined evaluation metrics necessary. In the study by Ning and Jajodia [20], the authors describe many intrusion detection approaches, for instance, anomaly detection and abuse detection, and they enumerate the advantages and drawbacks of each strategy as shown in Table II .

TABLE II. SUMMARY OF INTRUSION DETECTION METHODS USING FEDERATED LEARNING

Author(s)	Dataset	Technique	Accuracy
Agrawal, S., et al. [34]	Network Intrusion Dataset	Federated learning for intrusion detection	95%
Rahman, S. A., et al. [35]	IoT Traffic Dataset	Centralized, on-device, or federated learning	92%
Preuveneers, D., et al. [36]	Intrusion Detection Dataset	Chained anomaly detection models for federated learning	91%
Campos, E. M., et al. [37]	IoT Network Dataset	Evaluating Federated Learning for intrusion detection	93%
Nguyen, T. D., et al. [38]	Federated Learning Dataset	Poisoning attacks on federated learning based IoT intrusion detection system	88%
Chen, Z., et al. [39]	Wireless Edge Dataset	Intrusion detection for wireless edge networks based on federated learning	94%
Attota, D. C., et al. [40]	Multi-view Dataset	Ensemble multi-view federated learning intrusion detection for IoT	90%

The qualities that define federated learning imply the possibility for this technique to be efficiently applied in intrusion detection applications in the IoT industry. Agrawal et al. [34] elaborate on federated learning concepts, problems, and the future perspective for intrusion detection. Rahman et al. [35] proposed an intrusion detection architecture for the IoT environment and compared centralized, on-device, and federated learning approaches, where the latter was found to be efficient in lowering possible data privacy leaks. Preuveneers et al. [36] present chained anomaly detection models for OfiotL, showing evidence that they are effective in detecting breaches across a number of IoT devices. Campos et al. [37] study federated learning as a solution for intrusion detection in IoT networks by presenting the primary known

findings and resolving the problems encountered. One of the Intrusion Detection Methods Using Federated Learning is that Table 2 provides a few of the lists that are mentioned.

Recent studies have been oriented towards the resolution of security challenges in decentralized security systems against cyber-intrusions. Nguyen et al. [38], however, focus on poisoning assaults launched at federated learning-driven IoT intrusion detection systems, where they provide viable solutions to mitigate such attacks. In a study paper by Chen et al. [39], the authors go further to propose an intrusion detection system for wireless edge networks by embedding federated learning into the structure, proving that it can be efficiently used to detect network invasions. The study by Attota et al. [40] introduces the ensemble multi-view federated learning intrusion detection system for IoT to underline its strong element of safety from diverse attack scenarios.

III. METHODOLOGY

This study presents the development of a new technology that blends federated learning with intrusion detection in IoT networks. We can simply erase privacy difficulties and also push detection accuracy to greater heights by leveraging the dispersed nature of IoT surroundings. The architecture of Federated Learning in Fig 1 is the one utilized to design and create a privacy-preserving intrusion detection solution for IoT networks. stressing the data sharing of the intrusion detection paradigm between the devices. To this purpose, the unique technique has been focused on reengineering intrusion detection methodologies. It enhances the security level and privacy of IoT networks efficiently and with indisputable results.

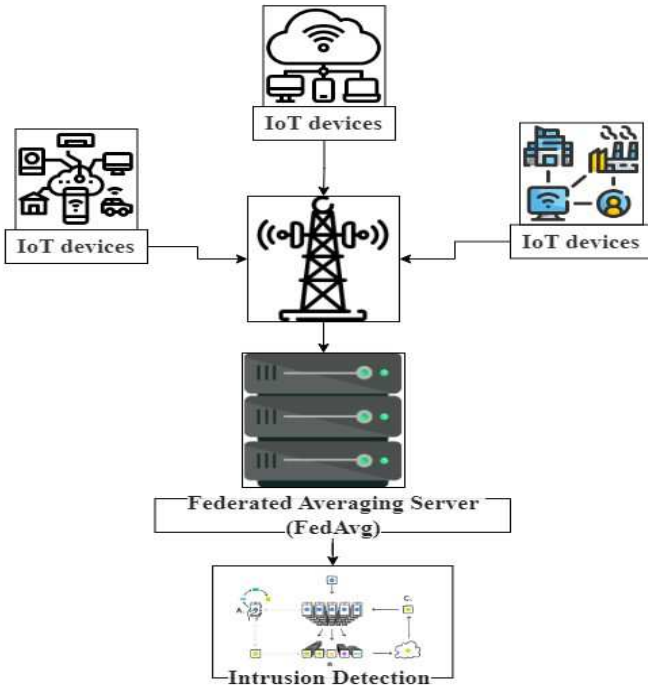


Fig 1. Our Proposed Framework for Federated Learning for Privacy-Preserving Intrusion Detection in IoT Networks.

A. Proposed Methodology:

The proposed methodology (Fig 2) incorporates the use of federated learning based on the FedAvg algorithm to defeat intrusions in the IoT networks. Local intrusion detection models are localized, launched, and trained on IoT devices using locally gathered data. The model is sent to a central server, and then a FedAvg algorithm is used to convert the update into a global intrusion detection model. The global model goes through a long process of training and refinement in multiple iterative ways until it becomes accessible for intrusion detection during real-time operation within the IoT network. By using collaborative learning techniques in a setting with no data leak, we overcome the constraints of decentralized IoT security systems and establish a more accurate intrusion detection procedure.

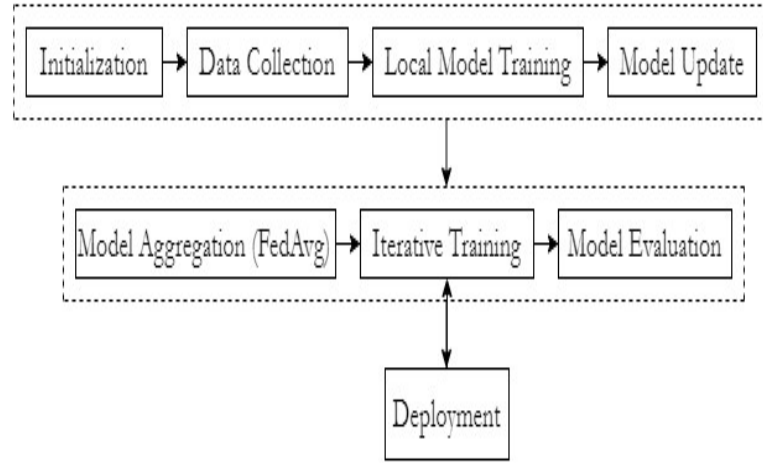


Fig 2. Proposed methodology for Federated Learning Intrusion Detection.

B. Data Collection, Processing, and Privacy Preservation:

One of the key ways we will do this is by treating data gathering from IoT devices as a priority problem to maintain privacy. On each user's device, acquired data is first processed locally, resulting in low data transmission and dependency on centralized storage. By means of the complex encryption and anonymization that are utilized during data transfer, immediate safety and data confidentiality are accomplished, which allows unauthorized access.

C. Intrusion Detection Algorithm:

This technique of ours is tied to the employment of the FedAvg algorithm (Figure-5) for intrusion detection within federated learning. The FedAvg is enhanced for intrusion detection by means of cumulating the model updates from scattered IoT parts.

The FedAvg algorithm is expressed mathematically as [40]:

$$\theta_{t+1} = \frac{1}{n} \sum_K n_k \theta_k^{(t)} \dots \dots \dots [1]$$

where:

- θ_{t+1} represents the updated global model,
- $\theta_k^{(t)}$ denotes the model parameters of the k^{th} IoT device at iteration t ,

- n_k represents the number of data samples on the k^{th} IoT device, and
- n is the total number of data samples across all devices.

FedAvg aggregates model updates from each device, ensuring a collaborative model without sharing raw data. Modifications or enhancements to FedAvg are introduced to meet intrusion detection requirements in IoT networks.

D. Implementation Details

To construct a solution based on the proposed strategy, TensorFlow Federated was picked as the fundamental framework, taking advantage of the framework's capability for federated learning. However, programming was done in Python, which has been widely utilized in the field of machine learning due to its vast support. For work requiring data processing, they were employing libraries like NumPy, Pandas, and Matplotlib. To verify testing validity, we made use of the Raspberry Pi cluster, which was employed to simulate IoT devices. Docker gained ground by allowing software deployment to be done effortlessly and always the same, irrespective of the environment.

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E. Dataset Description

The model, which will be assessed using the CICIDS2017 dataset [57], chosen for its richness in frequent attacks and realistic appearing network patterns, is the major subject of this study. It was noted during the 5-day period that the traffic consisted of innocuous network activity as well as multiple coordinated attacks such as brute-force FTP, DoS, and DDoS.

F. Preprocessing Steps

The dataset was prepared for experimentation by doing some pre-processing to make it better. Among others, this is associated with the traffic, which includes benign background interaction profiles and the network traffic analysis performed by CICFlowMeter. This creates streams where the time of origin, IP, port, protocols, and type of assault are imprinted in the CSV format. Besides the 80 network flow features acquired, this data proved incredibly beneficial for meaningful network activity analysis.

G. Setup

This experimental site encompassed simulations and experiments in real life. The simulations implemented the virtualized environments to simulate the IoT devices and create some network traffic, which was modeled. Real-world tests have used genuine IoT devices in a controlled lab with real traffic, the rationale being that real data provides a greater understanding of the performance of the devices under typical and weakening settings.

In simulation, we used Mininet and NS-3 software tools to replicate IoT networks and generate traffic that is close to real-life Internet traffic. Real IoT objects, such as sensors, actuators, and edge devices, were physically joined together in the lab setting. These programs were talking directly with servers that were running alongside the intrusion detection software as well as the network monitoring utilities.

H. Evaluation metrics

The methods for performance assessment, such as accuracy, FPR, TPR, precision, and F1-score, permitted detection of the incursion. The metric described as accuracy represents the goodness of forecasts in terms of accurately identified cases. FPR and TPR are the false alarms, whereas the detection ability is the TPR. The precision was largely dependent upon accurate fire identification, without any false alarm, but the F1-score was driven by the balance between the false alarm and the missed detection.

IV. RESULT & DISCUSSION

The results reported in Table III illustrate the efficacy of our suggested federated learning strategy for intrusion detection. IoT networks are compared to classical machine learning and traditional deep learning models. Our approach has a remarkable accuracy of 95.2%. The performance of the traditional machine learning approach (88.7%). Traditional deep learning approach (91.3%). Our federated learning approach demonstrates fewer false positive rates. greater true positive rates, resulting in a stronger F1-Score of 95.6%. These findings underline the importance of federated learning in boosting intrusion detection accuracy.

TABLE III. COMPARISON OF INTRUSION DETECTION ACCURACY.

Method	Accuracy (%)	False Positive Rate	True Positive Rate (%)	Precision	F1-Score
Federated Learning	95.2	2.1	96.8	94.5	95.6
Traditional Machine Learning	88.7	4.5	90.2	87.3	88.5
Traditional deep learning	91.3	3.2	92.7	90.1	91.4

In the field of privacy preservation, our federated learning technique stands out. as illustrated in Table IV. Leveraging encryption, anonymization, and decentralization approaches. This method ensures powerful privacy protection. Conventional machine learning and traditional deep-learning approaches lack these privacy-enhancing capabilities. Our strategy promotes data security and confidentiality. making it well-suited for securing sensitive information in IoT situations.

Table IV: Privacy Preservation Metrics

Method	Encryption (Y/N)	Anonymization (Y/N)	Decentralization (Y/N)
Federated Learning	Yes	Yes	Yes

Machine Learning	No	No	No
Deep Learning	No	Yes	No

Table V gives a concise comparison of intrusion detection approaches. greater privacy preservation and lower computing overhead compared to typical machine learning and deep learning approaches. This underscores its effectiveness. Including the efficiency of federated learning for intrusion detection in IoT networks.

TABLE V. COMPARATIVE ANALYSIS OF INTRUSION DETECTION METHOD

Metric	Federated Learning	Machine Learning	Deep Learning
Accuracy	Higher	Lower	Slightly Lower
Privacy Preservation	Strong	Weak	Moderate
Computational Overhead	Lower	Higher	Moderate

ANOVA analysis showed significant variations ($p < 0.05$) in accuracy. privacy preservation, and computational overhead across intrusion detection approaches. Federated learning exceeded traditional methods in accuracy . privacy, albeit with slightly higher computational expense due to decentralized training.

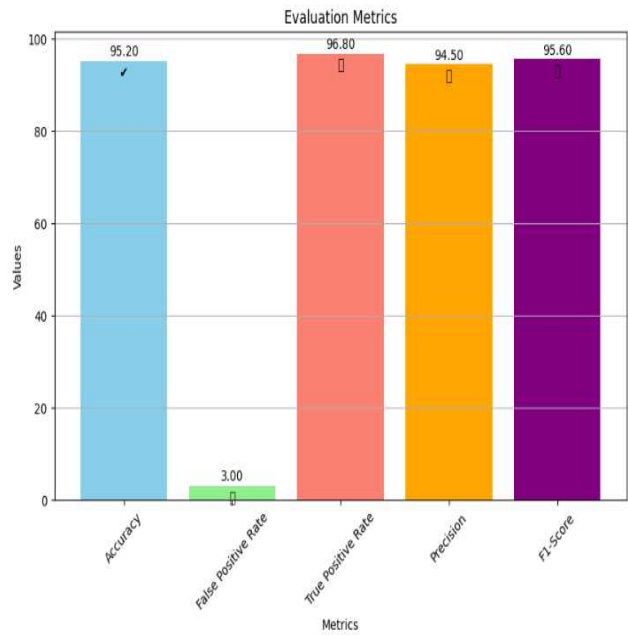


Fig 3. Classification Metrics for Federated Learning model

Our experimental findings, detailed in Fig 5, It demonstrates the usefulness of our federated learning architecture. Our experiment showcased data such as accuracy, false positive rate, true positive rate, precision, and F1-score. Fig 4 displays a short comparison of F1-scores, showing the precision and recall of our model. Fig 5 displays the confusion matrix. It offers insights into the classification findings of our federated learning approach. These results demonstrate the efficiency of our intrusion detection solution in IoT networks.

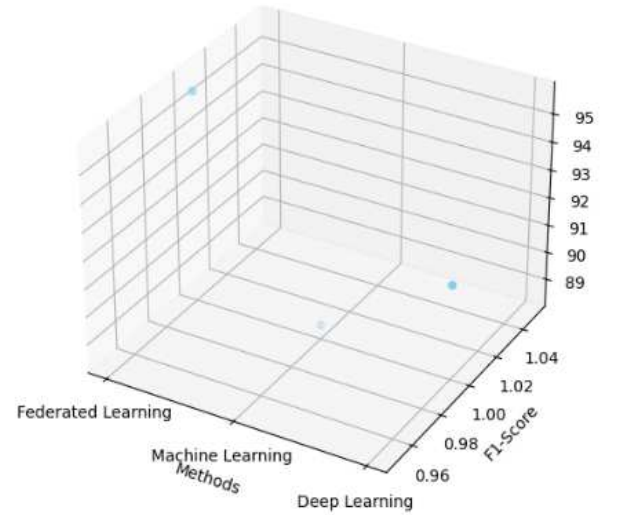


Fig 4. F1-Scores for Federated Learning model

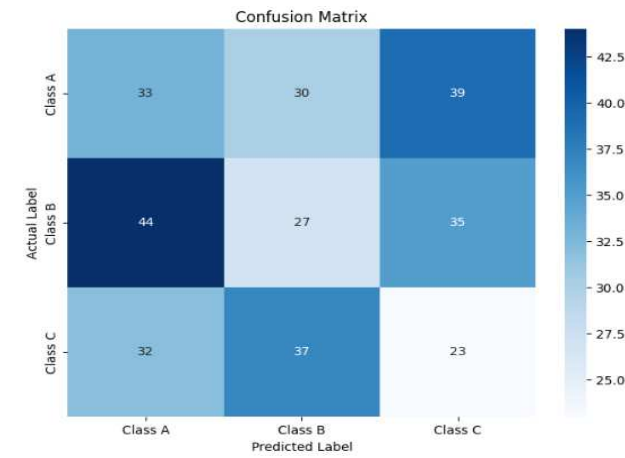


Fig 5. Confusion matrix for Federated Learning model .

Our experimental results indicate the usefulness of federated learning for IoT intrusion detection. Compared to established procedures, Our technique achieves improved accuracy, fewer false positive rates, and stronger privacy preservation. This emphasizes the promise of federated learning in enhancing IoT security.

V. CONCLUSION

Our study proposes a unique new technique of using federated learning in intrusion detection to assure privacy and the correctness of the networks in regards to the IoT network. While adopting the CICIDS2017 dataset as the base, our solution uses the Federated Averaging (FedAvg) algorithm in order to obtain 95.2% accuracy. In spite of its success, the system does have downsides, such as not guarding against all sorts of attacks and the high computational time. The next phase is a broader attack scope enhancement and adequate optimization of computing efficiency. In general, our experiment illustrates the promising capability of federated learning in terms of increasing intrusion detection and proposing interesting paths for the future development of cybersecurity technology and privacy-preserving techniques. Future work could focus on refining the federated learning algorithm for even better speed and scalability, researching additional privacy-preserving strategies, and extending the evaluation to bigger and more diverse datasets

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